SymbioForge

Heart-health decision support: eight AI workflows, one orchestrator, one explained overview

The problem

A bare yes/no from a single model is hard to trust and impossible to act on.

People do not know what their own numbers mean.

A health tool that cannot say "I am unsure" is unsafe.

Our answer

Run independent workflows (ML, deep learning, expert rules, SHAP, trust, what-if, test planner) and reconcile them.

Return an explained, uncertainty-aware overview and a next step, not a label.

Reach it through a clinic website: one link, one flow.

0.896

ROC-AUC, held-out test (n = 138)

10

AI workflows run and traced for every patient

56 ms

median time for the full pipeline incl. explanation

25

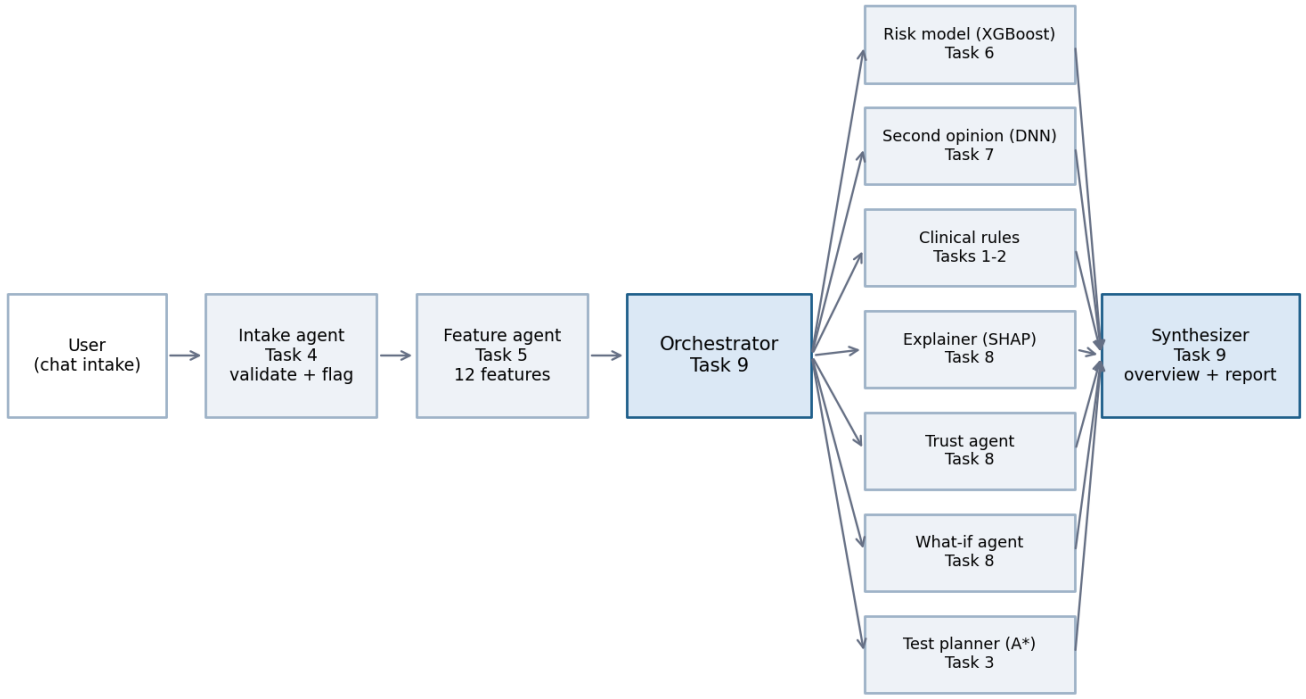
automated tests passing

What a visitor experiences



HOW IT WORKS

Eight workflows, one orchestrator



Every agent reports what it did; the orchestrator merges the evidence into one patient-level overview.

Workflow	From	What it does
Intake agent	T4	validate; flag and fill missing tests
Feature agent	T5	12 features from one formula module
Risk model	T6	XGBoost probability (the verdict)
Second opinion	T7	DNN; flags disagreement
Clinical rules	T1-2	R1-R7: fired / not fired / unknown
Test planner	T3	A* picks the cheapest decisive test
Explainer	T8	exact SHAP contributions
Trust + what-if	T8	bands, conformal set, sensitivity
Synthesizer	T9	overview, doctor questions, report

918 patients

4 hospitals, 918 after cleaning, 12 engineered features

Clean evaluation

70/15/15 split (seed 42); imputer, scaler and models fit on train only

No train/serve skew

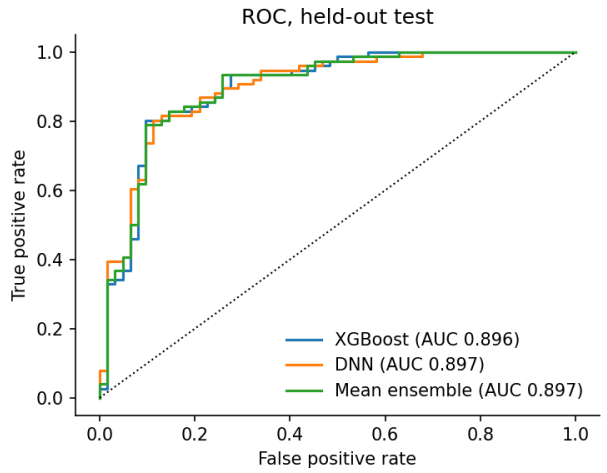
one feature function; parity test < 1e-9 against the saved matrix

Honest about gaps

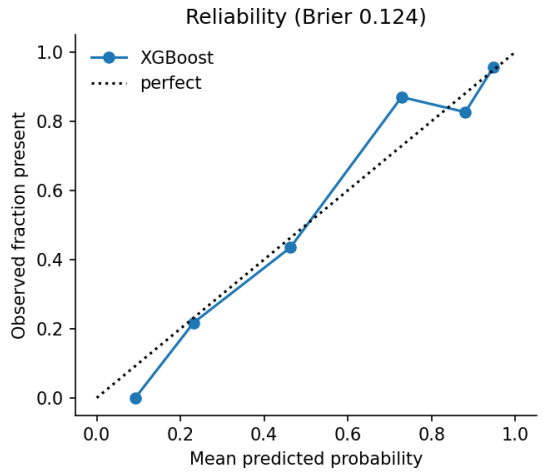
missing tests are flagged, confidence is capped, rules never run on guesses

EVIDENCE

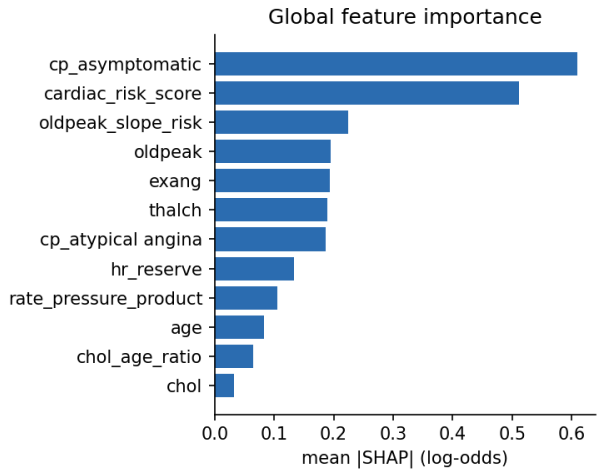
Accurate, and honest about uncertainty



ROC curves, held-out test (AUC XGBoost 0.896, DNN 0.897)



Reliability: predicted vs observed (Brier 0.124)



What drives predictions (mean |SHAP|, test set)

Seven models compared (5-fold CV on train, then held-out test)

Model	CV AUC	Test acc.	Test AUC
XGBoost (tuned)	0.876	0.819	0.896
LogisticRegression	0.874	0.833	0.904
SVM	0.859	0.819	0.879
RandomForest	0.853	0.841	0.870
GradientBoosting	0.853	0.833	0.893
XGBoost (task 6)	0.844	0.833	0.878
DecisionTree	0.811	0.819	0.855

92%
ML accuracy when rules agree

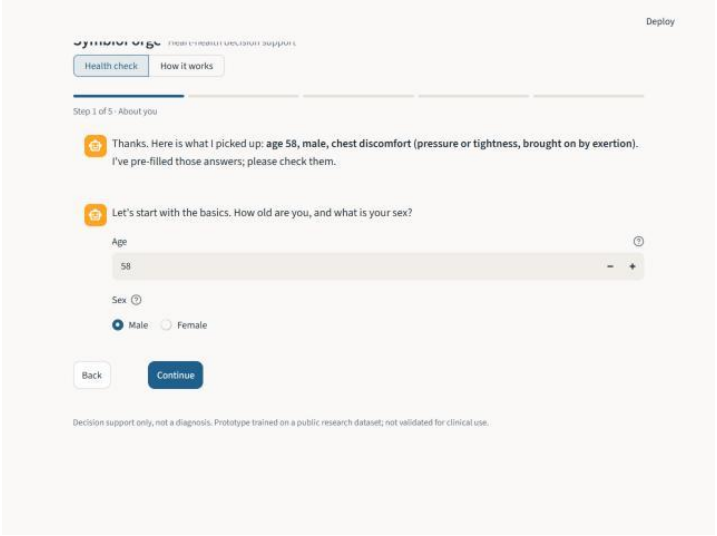
6%
XGBoost and DNN disagree; confidence is lowered

92%
conformal coverage (target 90%); 21% flagged 'uncertain'

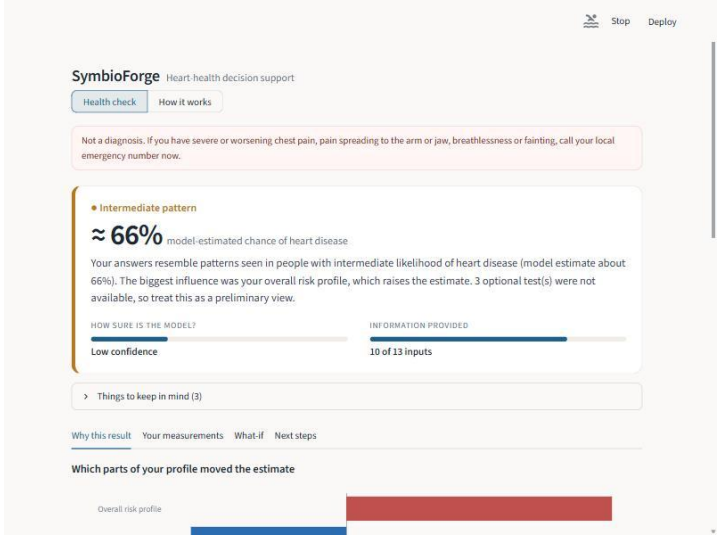
What we conclude

AUC 95% interval 0.84-0.95: XGBoost and the DNN cannot be separated at n = 138, and averaging gave no gain. So XGBoost gives the verdict, the DNN is a second opinion, and every disagreement is shown to the user.

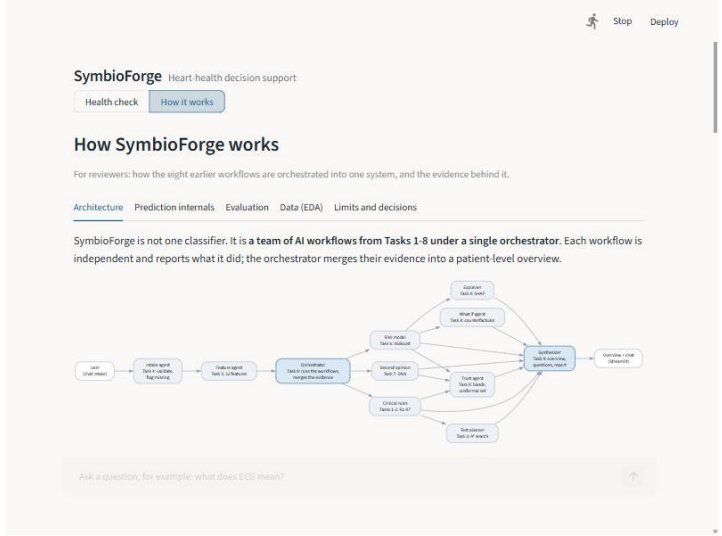
From a clinic website to a plain-language overview



Guided chat: plain-language questions, a tooltip on every field, free-text pre-fill



Overview: risk pattern, estimate, confidence and how much information was given



How it works view for reviewers: live trace of the workflows

What the user gets

Plain-language questions; chest pain classified from three yes/no questions.

Four tabs: why this result, their measurements vs reference ranges, what-if curves, next steps.

Three independent opinions (ML, deep learning, expert rules).

Ranked next tests, questions for the doctor, a downloadable report.

A chat that answers only from the computed analysis: no external AI service, no invented facts.

One public link, free to run

Clinic website (GitHub Pages) is the front door; its button redirects into the Streamlit Cloud app, which links back and offers booking.

Stateless: nothing stored, no API keys, no data leaves the page.

Local demo with one command: python run_demo.py

Live app: symbioforge.streamlit.app

What we guard against, what we learned, where it goes next

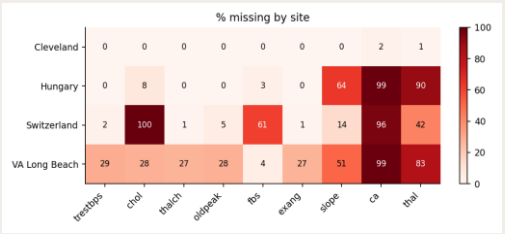
Area	How it is handled
Not a diagnosis	Notice on every screen and report; emergency wording always available
Missing tests	Flagged, never silent; 2+ missing blocks 'high' confidence; rules never run on guesses
Disagreement	ML vs DNN vs rules shown; any conflict lowers confidence
Explainable	Exact SHAP per patient; DNN limits stated in the UI
Uncertainty	Validation-based bands and a 90% conformal set; 'uncertain' is allowed
Counterfactuals	Labelled model sensitivity, never treatment advice
Privacy	Stateless; no patient data stored or sent anywhere

Fairness check (test set; groups are small, so weak evidence)

Group	n	AUC	Recall
Sex: male	113	0.881	0.843
Sex: female	25	0.895	0.833
Site: Cleveland	46	0.898	0.783
Site: Hungary	53	0.897	0.800
Site: Switzerland	18	0.941	0.882
Site: VA Long Beach	21	0.825	0.938

What independent review caught

Two review passes found defects that tests had missed. The most telling: a constant used to fill missing cholesterol acted as a hidden hospital marker, and the model learned it. Removing it lowered test AUC slightly (0.899 to 0.896), which is the honest number.



Why we are careful

Missingness and disease rate both depend on the hospital, so the data hide a confounder.

Conclusion

SymbioForge turns a classifier into an explained, uncertainty-aware overview by reconciling eight workflows, and it is reached through a clinic front door with a single public link.

Honest limits: small test set (138), historical single-source data, no clinical validation.

Next: clinician and user study, site-held-out validation, calibrated probabilities, an optional language-model layer limited to computed evidence.